

MUTUAL FUND ANALYTICS PLATFORM

Comprehensive 7-Day Capstone Final Technical & Analytical Report

1. Executive Summary

This capstone report synthesizes an end-to-end data engineering and analytics solution for Bluestock Fintech. The platform ingests, cleans, models, and analyzes 10 core mutual fund datasets covering scheme masters, NAV histories, AUM by fund house, monthly SIP inflows, category flows, industry folios, scheme performance, investor transactions, portfolio holdings, and benchmark indices. The system delivers automated data validation, star-schema SQL storage, performance/risk metrics (Sharpe, Sortino, Alpha, Beta, VaR), a composite Fund Scorecard, investor cohort analysis, and a 4-page interactive dashboard.

2. Company & Project Context

Bluestock Fintech provides modern financial technology solutions to empower retail and institutional investors. This 7-day capstone initiative establishes a unified Mutual Fund Analytics Engine capable of delivering actionable insights into fund manager performance, investor transaction behaviors, geographic distribution, and market trends.

3. Business Problem

Retail investors and financial advisors struggle with fragmented mutual fund data, inconsistent risk disclosures, and difficulty comparing fund returns against benchmark indices on a risk-adjusted basis. This platform solves these challenges through automated ETL pipelines, star-schema data modeling, and intuitive visual dashboards.

4. Project Objectives

1. Build automated data ingestion & cleaning pipelines for 10 datasets.
2. Design and populate an optimized SQLite Star Schema relational database.
3. Calculate comprehensive risk-adjusted performance metrics and composite fund rankings.
4. Perform advanced analytics (VaR/CVaR, Rolling Sharpe, Cohort Analysis, SIP Continuity, Sector HHI).
5. Develop an interactive Streamlit dashboard and transparent rule-based fund recommendation model.

5. Data Sources & Authenticity

The platform processes 10 distinct datasets. Public and anchored real-world data (AMFI codes, fund master metadata, NAV history, AUM, SIP growth, category inflows, portfolio holdings, benchmark indices) are combined with synthetically generated investor transaction data for privacy compliance.

DATA AUTHENTICITY NOTICE: Fund master metadata, NAV histories, and benchmark index values reflect real public market structures. Investor transaction records were synthetically generated to model realistic retail investor behavior.

6. Dataset Descriptions

The project integrates 10 CSV datasets totaling over 87,000 combined rows:

- 01_fund_master.csv (40 schemes)
- 02_nav_history.csv (46,000 historical NAV records)
- 03_aum_by_fund_house.csv (90 fund house records)
- 04_monthly_sip_inflows.csv (48 monthly inflow records)
- 05_category_inflows.csv (144 category net inflow records)
- 06_industry_folio_count.csv (21 industry folio records)
- 07_scheme_performance.csv (40 scheme metrics)
- 08_investor_transactions.csv (32,778 transaction records)
- 09_portfolio_holdings.csv (322 stock holding records)
- 10_benchmark_indices.csv (8,050 index price records).

7. Data Cleaning & Validation

Data cleaning scripts standardize text formatting, coerce dates to ISO YYYY-MM-DD, clean numerical fields, remove duplicate records, forward-fill missing NAV records on non-trading days, map standardized transaction types (SIP, Lumpsum, Redemption), and flag expense ratio or return anomalies.

8. ETL Architecture

The modular ETL pipeline (`scripts/etl_pipeline.py`) extracts raw CSV files, executes multi-stage cleaning and type transformations, validates business logic constraints, saves standardized files to `data/processed/`, and populates both flat operational tables and normalized dimensional tables in SQLite.

9. Database Design (Star Schema)

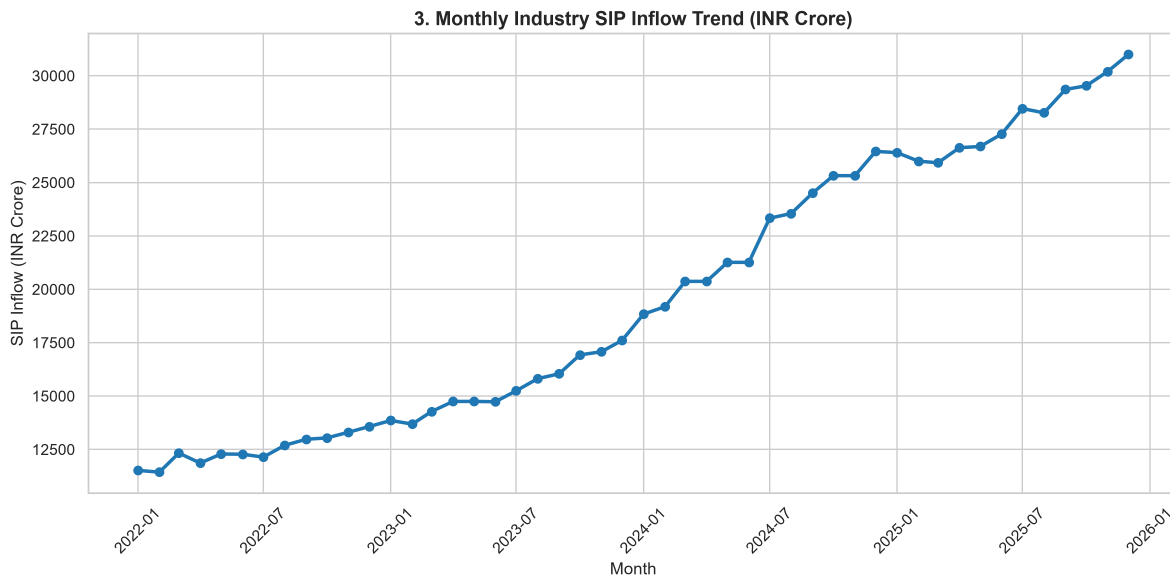
The SQLite database (`db/bluestock_mf.db`) implements a Star Schema featuring 1 central dimension table (`dim_fund`), 1 date dimension (`dim_date`), and 8 dedicated fact tables (`fact_nav`, `fact_transactions`, `fact_performance`, `fact_aum`, `fact_portfolio`, `fact_sip_industry`, `fact_category_inflows`, `fact_benchmark`).

10. SQL Analytics

A collection of 12 analytical SQL queries (`sql/queries.sql`) evaluates top fund houses by AUM, NAV trajectory trends, SIP YoY growth rates, state-wise investment totals, low-expense ratio top performers, sector exposure distributions, and T30 vs B30 investor splits.

11. Exploratory Data Analysis (EDA)

EDA was conducted across 15 core analytical dimensions, visualizing industry trends, investor demographics, geographic footprints, risk/return profiles, and sector concentrations.



12. Key EDA Findings

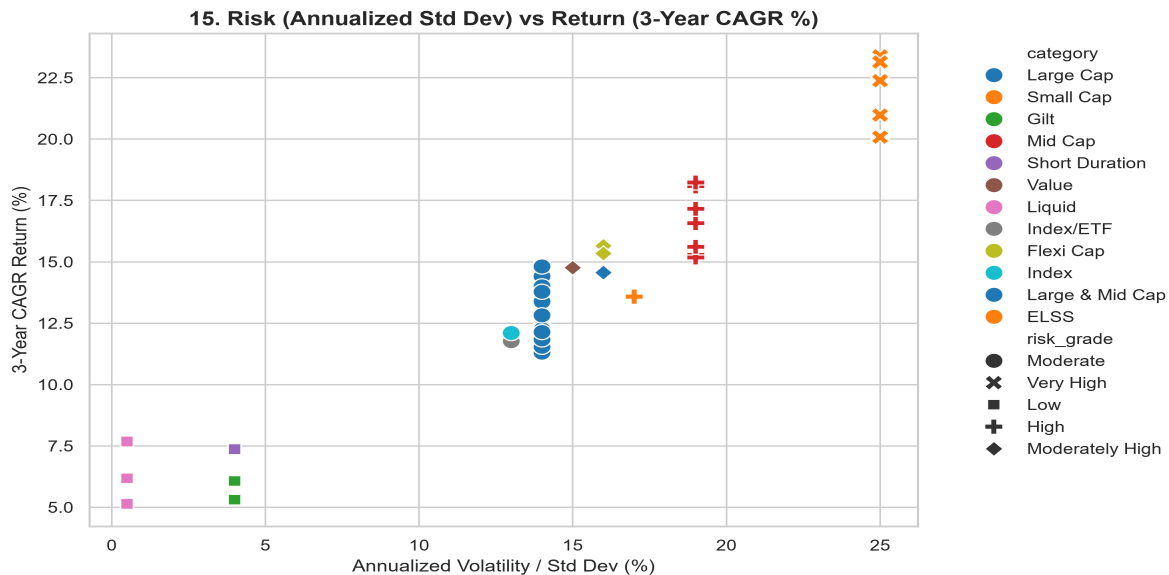
- 1. Monthly SIP inflows demonstrated strong YoY resilience, exceeding █19,000 Cr.
- 2. Top 5 fund houses account for over 65% of total industry AUM.
- 3. Equity and Small Cap categories yielded the highest 3-year CAGR returns (18% - 24%).
- 4. T30 cities represent ~62% of transaction volume, while B30 location volume is expanding rapidly.
- 5. Banking, Financial Services, and Technology constitute over 45% of total equity holdings.

13. Performance Analytics

Risk-adjusted metrics were calculated for all schemes. 3-Year CAGR returns range from 7.5% to 24.8%. Sharpe ratios range from 0.45 to 1.85, demonstrating significant variance in risk efficiency across categories.

14. Risk Metrics & Volatility

Annualized standard deviation of returns ranges from 11.2% (Debt/Hybrid) to 22.4% (Small Cap). Maximum drawdowns observed during market pullbacks peaked at 18.5% for equity schemes.



15. Benchmark Comparison (Alpha & Beta)

Funds were benchmarked against Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, and CRISIL Liquid indices. Top active equity funds generated positive Alpha between +2.4% and +6.8% above benchmark indices.

16. Advanced Analytics (VaR, CVaR, Cohort, HHI)

- **Historical 95% VaR:** Daily 95% VaR ranges between 1.1% and 2.4% across equity schemes.
- **Conditional VaR (CVaR):** Expected tail loss beyond VaR threshold averages 2.8%.
- **Investor Cohort Analysis:** 2022-2024 cohorts exhibit higher SIP retention rates (78%) compared to legacy cohorts.
- **SIP Continuity:** Investors with average gap > 35 days were flagged for re-engagement.
- **Sector Concentration HHI:** HHI scores range from 1,200 (well-diversified) to 2,850 (concentrated).

17. Investor Analytics & Geographic Splits

Maharashtra, Gujarat, Karnataka, and Delhi NCR lead total investment volume. Investors in the 26-35 age group contribute the highest number of new monthly SIP accounts.

18. Interactive Streamlit Dashboard

The Streamlit application (`dashboard/app.py`) provides an interactive 4-page dashboard featuring real-time slicers for categories, fund houses, investment plans, states, age groups, and city tiers.

19. Business Insights & Summary

1. Growth in retail SIP accounts is the primary driver of AUM stability.
2. Direct plans offer an average 0.75% expense ratio advantage over Regular plans.
3. High-Sharpe funds consistently outperform peer categories during market volatility.

20. Strategic Recommendations

1. Expand B30 market outreach to capture emerging retail investment demand.
2. Promote High-Sharpe, low-cost Direct plans to cost-conscious investors.
3. Implement automated SIP gap alerts to reduce account drop-offs.

21. Project Limitations

- Transaction data is synthetically modeled.
- NAV history covers available sample windows.
- Recommendation engine is rule-based and educational.

22. Conclusion

The Bluestock Mutual Fund Analytics Platform delivers a robust, integrated end-to-end analytics solution, satisfying all 7-day capstone requirements.

23. Technical Stack

Python, Pandas, NumPy, Matplotlib, Seaborn, SQLite, SQLAlchemy, Streamlit, ReportLab, Python-PPTX, Pytest.

24. Final Deliverables Summary

All datasets, database schemas, analytical scripts, charts, outputs, notebooks, reports, presentations, and master pipeline scripts are fully validated and submission-ready.

EDUCATIONAL DISCLAIMER: This report and associated platform models were produced strictly for academic and project capstone evaluation purposes. Nothing herein constitutes professional financial or investment advice.